Computer Vision

Homework 3: Tracking

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Based on Assignment 3 of CS 585 (Spring 2024), a programming assignment on tracking. Teams of up to three are fine. Goes with Chapter 3 of the notes.

The goal is to implement a single-object tracking filter and a multi-object data association algorithm.

Data. The course provided detections from a traffic video. If you don't have that data, record a short video of moving objects (cars from a window, people in a hallway, balls on a table), detect the objects in each frame with the methods of Chapter 1, and save the detections as one list of centres or bounding boxes per frame. The synthetic generator in Section 3.2.2 of the notes is a useful first test.

1. **One object.** Given the noisy detected 2D positions of one vehicle over the frames of a video, implement an alpha-beta filter or a Kalman filter (Sections 3.1.2 and 3.1.3) and produce a smooth track. Show the measurements and the filtered track on the frames, explain how you chose the parameters, and show what happens when some detections are missing.
2. **Many objects.** Given bounding boxes of several objects in every frame, track all of them and give each object an ID that stays the same for as long as it is detected. Use a prediction for every track, a cost matrix and an assignment (greedy or optimal), as in Section 3.2.
3. **Questions.** Answer the following in your report, with examples from your results.
   (a) Show an easy part of the sequence, a hard part where your tracker succeeds, and a hard part where it fails.
   (b) How do you decide when to start a new track and when to end an old one, as objects enter and leave the view?
   (c) What happens when two objects touch or occlude each other, and how could you keep their tracks from breaking or swapping?
   (d) What happens with spurious detections that do not continue in later frames?
   (e) What are the advantages and drawbacks of different motion models (constant position, constant velocity, constant acceleration)?

Hand in your code, a report (PDF) with figures of the tracks drawn on the frames, one colour per ID, and a short video of the result.